

Yumeng He

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Salt Lake City, UT

Education

- **University of Utah** August 2024 - Present
PhD in Computer Science
Salt Lake City, UT
 - GPA: 4.00/4.00, advised by Pavel Panchekha
 - Research Interests: Program analysis, numerical computing, and formal verification.
- **University of Rochester** August 2020 - May 2024
Bachelor of Science, Double Major in Computer Science and Mathematics, Minor in Business
Rochester, NY
 - Grade: 3.98/4.00, graduated summa cum laude, highest distinction.

Publications

- [1] He, Y., & Panchekha, P. (2026). Accurate Residues for Floating-Point Debugging. OOPSLA 2026 (To Appear).
- [2] He, Y., Nandi, C., & Pai, S. (2025). Formalizing Linear-Motion G-code for Invariant Checking and Differential Testing of Fabrication Tools. OOPSLA 2025. [\[DOI\]](#) [\[Talk\]](#) [\[Code\]](#)

Professional Experience

- **ML Compiler Engineer Intern** May 2026 - Aug 2026
Annapurna Labs, AWS
Cupertino, CA
 - Developed compiler support for dynamic tensor shapes on AWS Trainium and Inferentia accelerators.
 - Designed and implemented an LICM pass with SSA-based safety and register-pressure checks.
 - Extended existing IR, register-allocation, and optimization passes from general-purpose to hardware shape registers.

Projects

- **Math Library Verification** August 2026 - Present
 - Developed a verification framework that derives sound, tight error bounds for math library implementations.
- **Efficient Numeric Debugging with Improved Accuracy [1]** August 2024 - May 2026
 - Improved diagnostic accuracy in an LLVM-based floating-point debugger without using high-precision arithmetic.
 - Extended a WebAssembly interpreter for shadow execution to track per-variable floating-point error.
 - Evaluated on 44 real-world workloads, showing higher accuracy on cases where baseline systems yield false results.
- **Differential Fuzzing for Bug Detection in Fabrication Tools [2]** May 2023 - October 2025
 - Developed a static analysis tool to detect software bugs in 3D printing pipelines using differential testing on G-code.
 - Defined formal G-code semantics for scalable geometric comparison.
 - Detected rotation-invariant violations across 58 real-world models, revealing bugs in production fabrication tools.
- **Lease Cache Design** May 2023 - August 2023
 - Modeled performance degradation resulting from optimal virtual lease assignments in physical cache systems.
 - Developed mathematical proofs for key properties of the model.

Honors and Awards

- **Excellence in Undergraduate Research** May 2024
Department of Computer Science, University of Rochester
- **Phi Beta Kappa** March 2023
University of Rochester Chapter

Academic Service

- POPL 2027 Artifact Evaluation Committee Member
- MFCS 2024 Referee

Miscellaneous

- TA for Vibe Coding, CS Dept., University of Utah Spring 2026
- TA for Computer Architecture, CS Dept., University of Rochester Spring 2023
- TA for Operations Research, Math Dept., University of Rochester Fall 2022